

Mehdi B. Tahoori

professor

基本信息

姓名: Mehdi B. Tahoori 所属院校: 卡尔斯鲁厄理工学院
邮箱: mehdi.tahoori@kit.edu 地区: Germany
年龄范围: 40岁至50岁 海外经历: 有
企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

电子与计算机工程

集成电路设计与集成系统

微电子科学与工程

信息安全

研究领域

计算机体系结构

低功耗电路与能量管理

低功耗设计与能耗优化

集成电路设计与制造

电路与系统设计

印刷电子学与可拉伸电路

系统安全与漏洞分析

EDA工具与电路验证

深度学习与神经网络

微电子器件与制造

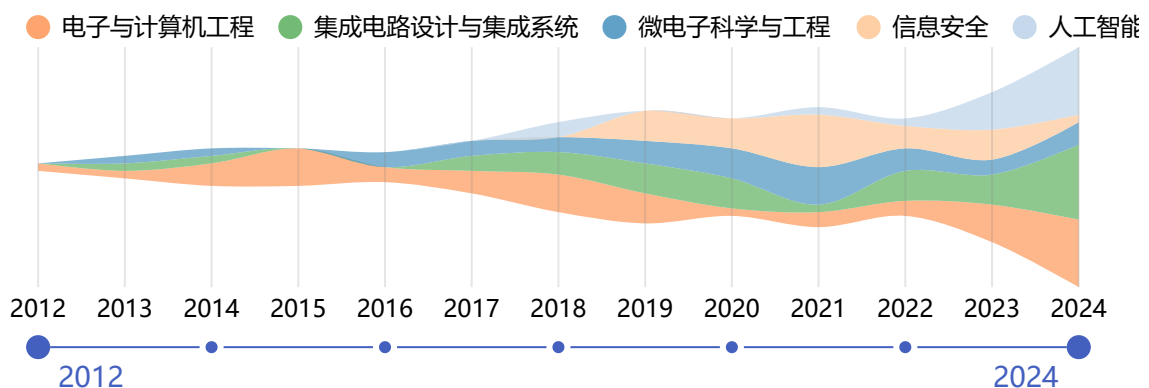
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个人简介

Prof. Dr. Mehdi B. Tahoori is a full professor and Chair of Dependable Nano - Computing at KIT, Germany. He holds a PhD and M.S. from Stanford University

and a B.S. from Sharif University of Technology. He has worked at Northeastern University, Fujitsu, etc. He has over 400 publications and holds multiple patents. His research focuses on reliable, secure, and energy - efficient computing. He serves as deputy editor - in - chief of IEEE Design and Test Magazine and has held various editorial positions. He chaired multiple IEEE symposia. He has received many awards, including NSF CAREER Award and ERC Advanced Grant, and is an IEEE fellow. Office hours are Wednesdays, 13:00 - 14:00, room B2 - 307.1, phone +91 721 608 47778, fax +91 721 608 43962.

近年发表的论文



Hardware and Software Co-Design for Optimized Decoding Schemes and Application Mapping in NVM Compute-in-Memory Architectures

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Siddaramu, Shanmukha Mangadahalli; Nezhadi, Ali; Mayahinia, Mahta; Ghasemi, Seyedehmaryam; Tahoori, Mehdi B.

被引用次数: 0

2024

One-Shot Online Testing of Deep Neural Networks Based on Distribution Shift Detection

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ahmed, Soayed Tuhin; Tahoori, Mehdi B.

被引用次数: 0

2024

Improved Arithmetic Performance by Combining Stateful and Non-Stateful Logic in Resistive Random Access Memory 1T-1R Crossbars

ADVANCED INTELLIGENT SYSTEMS

Brackmann, Leon; Ziegler, Tobias; Jafari, Atousa; Wouters, Dirk J.; Tahoori, Mehdi B.; Menzel, Stephan

被引用次数: 2

2024

Design-time Reference Current Generation for Robust Spintronic-based Neuromorphic Architecture

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Ahmed, Soyed Tuhin; Mayahinia, Mahta; Hefenbrock, Michael; Muench, Christopher; Tahoori, Mehdi B.
被引用次数: 1 2024

Ultra-Scaled E-Tree-Based SRAM Design and Optimization With Interconnect Focus

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Pei, Zhenlin; Liu, Hsiao-Hsuan; Mayahinia, Mahta; Tahoori, Mehdi B.; Catthoor, Francky; Tokei, Zsolt; Abdi, Dawit Burusie; Myers, James; Pan, Chenyun

被引用次数: 0 2024

Soft and Hard Error-Correction Techniques in STT-MRAM

IEEE DESIGN & TEST

Hemaram, Surendra; Tahoori, Mehdi B.; Catthoor, Francky; Rao, Siddharth; Couet, Sebastien; Pica, Valerio; Kar, Gouri Sankar

被引用次数: 0 2024

Dynamic Segmented Bus for Energy-Efficient Last-Level Cache in Advanced Interconnect-Dominant Nodes

IEEE EMBEDDED SYSTEMS LETTERS

Mayahinia, Mahta; Marinelli, Tommaso; Pei, Zhenlin; Liu, Hsiao-Hsuan; Pan, Chenyun; Tokei, Zsolt; Catthoor, Francky; Tahoori, Mehdi B.

被引用次数: 0 2024

Reducing ADC Front-End Costs During Training of On-Sensor Printed Multilayer Perceptrons

IEEE EMBEDDED SYSTEMS LETTERS

Afentaki, Florentia; Duarte, Paula Carolina Lozano; Zervakis, Georgios; Tahoori, Mehdi B.

被引用次数: 0 2024

Embedding Hardware Approximations in Discrete Genetic-based Training for Printed MLPs

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Afentaki, Florentia; Hefenbrock, Michael; Zervakis, Georgios; Tahoori, Mehdi B.

被引用次数: 0 2024

Enhancing Reliability of Neural Networks at the Edge: Inverted Normalization with Stochastic Affine Transformations

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Ahmed, Soyed Tuhin; Danouchi, Kamal; Prenat, Guillaume; Anghel, Lorena; Tahoori, Mehdi B.

被引用次数: 0 2024

NeuSpin: Design of a Reliable Edge Neuromorphic System Based on Spintronics for Green AI

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Ahmed, Soyed Tuhin; Danouchi, Kamal; Prenatt, Guillaume; Anghel, Lorena; Tahoori, Mehdi B.

被引用次数: 0 2024

On-sensor Printed Machine Learning Classification via Bespoke ADC and Decision Tree Co-Design

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Armeniakos, Giorgos; Duarte, Paula L.; Pal, Priyanjana; Zervakis, Georgios; Tahoori, Mehdi B.; Soudris, Dimitrios

被引用次数: 0

2024

DropHD: Technology/Algorithm Co-design for Reliable Energy-efficient NVM-based Hyperdimensional Computing under Voltage Scaling

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Genssler, Paul R.; Mayahiniat, Mahta; Thomann, Simon; Tahoori, Mehdi B.; Amrouch, Hussam

被引用次数: 0

2024

In-field Detection of Small Delay Defects and Runtime Degradation using On-Chip Sensors

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Ghasemi, S. Maryam; Meschkov, Sergej; Krautter, Jonas; Gnad, Dennis R. E.; Tahoori, Mehdi B.

被引用次数: 0

2024

Algorithm to Technology Co-Optimization for CiM-based Hyperdimensional Computing

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Mayahinia, Mahta; Thomann, Simon; Genssler, Paul R.; Muench, Christopher; Amrouch, Hussam; Tahoori, Mehdi B.

被引用次数: 0

2024

Analog Printed Spiking Neuromorphic Circuit

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Pal, Priyanjana; Zhao, Haibin; Shatta, Maha; Hefenbrock, Michael; Mamaghani, Sina Bakhtavari; Nassif, Sani; Beigl, Michael; Tahoori, Mehdi B.

被引用次数: 0

2024

Side-Channel Attack with Fault Analysis on Memristor-based Computation-in-Memory

2024 IEEE 30TH INTERNATIONAL SYMPOSIUM ON ON-LINE TESTING AND ROBUST SYSTEM DESIGN, IOLTS 2024

Sapui, Brojogopal; Meschkov, Sergej; Tahoori, Mehdi B.

被引用次数: 0

2024

NN-ECC: Embedding Error Correction Codes in Neural Network Weight Memories using Multi-task Learning

2024 IEEE 42ND VLSI TEST SYMPOSIUM, VTS 2024

Ahmed, Soyed Tuhin; Hemaram, Surendra; Tahoori, Mehdi B.

被引用次数: 2

2024

Testing Spintronics Implemented Monte Carlo Dropout-Based Bayesian Neural Networks

IEEE EUROPEAN TEST SYMPOSIUM, ETS 2024

Ahmed, Soyed Tuhin; Danouchi, Kamal; Hefenbrock, Michael; Prenat, Guillaume; Anghel, Lorena; Tahoori, Mehdi B.

被引用次数: 0

2024

Degradation Monitoring Through Software-controlled On-chip Sensors for RISC-V

IEEE EUROPEAN TEST SYMPOSIUM, ETS 2024

Ghasemi, S. Maryam; Krautter, Jonas; Gheshlaghi, Tara; Meschkov, Sergej; Gnad, Dennis R. E.; Tahoori, Mehdi B.

被引用次数: 0

2024

Hard Error Correction in STT-MRAM

29TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2024

Hemaram, Surendra; Tahoori, Mehdi B.; Catthoor, Francky; Rao, Siddharth; Couet, Sebastien; Kar, Gouri Sankar

被引用次数: 0

2024

Power Side-Channel Analysis and Mitigation for Neural Network Accelerators based on Memristive Crossbars

29TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2024

Sapui, Brojogopal; Tahoori, Mehdi B.

被引用次数: 0

2024

A Visionary Look at the Security of Reconfigurable Cloud Computing

PROCEEDINGS OF THE IEEE

Stojilovic, Mirjana; Rasmussen, Kasper; Regazzoni, Francesco; Tahoori, Mehdi B.; Tessier, Russell

被引用次数: 0

2023

Co-Design of Approximate Multilayer Perceptron for Ultra-Resource Constrained Printed Circuits

IEEE TRANSACTIONS ON COMPUTERS

Armeniakos, Giorgos; Zervakis, Georgios; Soudris, Dimitrios; Tahoori, Mehdi B.; Henkel, Jorg

被引用次数: 3

2023

New Approaches of Side-Channel Attacks Based on Chip Testing Methods

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Meschkov, Sergej; Gnad, Dennis R. E.; Krautter, Jonas; Tahoori, Mehdi B.

被引用次数: 3

2023

NeuroScrub plus : Mitigating Retention Faults Using Flexible Approximate Scrubbing in Neuromorphic Fabric Based on Resistive Memories

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ahmed, Soyed Tuhin; Hefenbrock, Michael; Muench, Christopher; Tahoori, Mehdi B.

被引用次数: 2

2023

SpinBayes: Algorithm-Hardware Co-Design for Uncertainty Estimation Using Bayesian In-Memory Approximation on Spintronic-Based Architectures

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Ahmed, Soyed Tuhin; Danouchi, Kamal; Hefenbrock, Michael; Prenat, Guillaume; Anghel, Lorena; Tahoori, Mehdi B.

被引用次数: 2

2023

SpinDrop: Dropout-Based Bayesian Binary Neural Networks With Spintronic Implementation

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Ahmed, Soyed Tuhin; Danouchi, Kamal; Munch, Christopher; Prenat, Guillaume; Anghel, Lorena; Tahoori, Mehdi B.

被引用次数: 11

2023

Fault-Tolerant Neuromorphic Computing With Memristors Using Functional ATPG for Efficient Recalibration

IEEE DESIGN & TEST

Ahmed, Soyed Tuhin; Tahoori, Mehdi B.

被引用次数: 0

2023

Highly-dependable printed neuromorphic circuits based on additive manufacturing

FLEXIBLE AND PRINTED ELECTRONICS

Zhao, Haibin; Hefenbrock, Michael; Beigl, Michael; Tahoori, Mehdi B.

被引用次数: 2

2023

Power-Aware Training for Energy-Efficient Printed Neuromorphic Circuits

2023 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Zhao, Haibin; Pal, Priyanjana; Hefenbrock, Michael; Beigl, Michael; Tahoori, Mehdi B.

被引用次数: 2

2023

Scalable Spintronics-based Bayesian Neural Network for Uncertainty Estimation

2023 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Ahmed, Soyed Tuhin; Danouchi, Kamal; Hefenbrock, Michael; Prenat, Guillaume; Anghel, Lorena; Tahoori, Mehdi B.

被引用次数: 3

2023

Hardware-Aware Automated Neural Minimization for Printed Multilayer Perceptrons

2023 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Kokkinis, Argyris; Zervakis, Georgios; Siozios, Kostas; Tahoori, Mehdi B.; Henkel, Jorg

被引用次数: 0

2023

Split Additive Manufacturing for Printed Neuromorphic Circuits

2023 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Zhao, Haibin; Hefenbrock, Michael; Beigl, Michael; Tahoori, Mehdi B.

被引用次数: 0

2023

Highly-Bespoke Robust Printed Neuromorphic Circuits

2023 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Zhao, Haibin; Sapui, Brojogopal; Hefenbrock, Michael; Yang, Zhidong; Beigl, Michael; Tahoori, Mehdi B.

被引用次数: 2

2023

Security Challenges and Opportunities of Cloud FPGAs

PROCEEDINGS OF THE 2023 CLOUD COMPUTING SECURITY WORKSHOP, CCSW 2023

Tahoori, Mehdi B.

被引用次数: 0

2023

Bespoke Approximation of Multiplication-Accumulation and Activation Targeting Printed Multilayer Perceptrons

2023 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Afentaki, Florentia; Saglam, Gurol; Kokkinis, Argyris; Siozios, Kostas; Zervakis, Georgios; Tahoori, Mehdi B.

被引用次数: 2

2023

Automated Masking of FPGA-Mapped Designs

2023 33RD INTERNATIONAL CONFERENCE ON FIELD-PROGRAMMABLE LOGIC AND APPLICATIONS, FPL

Mueller, Nicolai; Meschkov, Sergej; Gnad, Dennis R. E.; Tahoori, Mehdi B.; Moradi, Amir

被引用次数: 0

2023

Fast and Efficient High-Sigma Yield Analysis and Optimization Using Kernel Density Estimation on a Bayesian Optimized Failure Rate Model

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Weller, Dennis D.; Hefenbrock, Michael; Beigl, Michael; Tahoori, Mehdi B.

被引用次数: 8

2022

Counterfeit Detection and Prevention in Additive Manufacturing Based on Unique Identification of Optical Fingerprints of Printed Structures

IEEE ACCESS

Erozan, Ahmet Turan; Hefenbrock, Michael; Gnad, Dennis R. E.; Beigl, Michael; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 1

2022

Remote Fault Attacks in Multitenant Cloud FPGAs

IEEE DESIGN & TEST

Krautter, Jonas; Gnad, Dennis R. E.; Tahoori, Mehdi B.

被引用次数: 2

2022

Artificial Neurons on Flexible Substrates: A Fully Printed Approach for Neuromorphic Sensing

SENSORS

Singaraju, Surya A.; Weller, Dennis D.; Gspann, Thuriid S.; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 4

2022

Data Leakage through Self-Terminated Write Schemes in Memristive Caches

27TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2022

Krautter, Jonas; Mayahinia, Mahta; Gnad, Dennis R. E.; Tahoori, Mehdi B.

被引用次数: 2

2022

Adaptive Block Error Correction for Memristive Crossbars

2022 IEEE 28TH INTERNATIONAL SYMPOSIUM ON ON-LINE TESTING AND ROBUST SYSTEM DESIGN (IOLTS 2022)

Hemaram, Surendra; Mayahinia, Mahta; Tahoori, Mehdi B.

被引用次数: 3

2022

Breaking an FPGA-integrated NIST SP 800-193 compliant TRNG Hard-IP core with on-chip voltage-based fault attacks

2022 32ND INTERNATIONAL CONFERENCE ON FIELD-PROGRAMMABLE LOGIC AND APPLICATIONS, FPL
Gnad, Dennis R. E.; Hu, Jiaqi; Tahoori, Mehdi B.

被引用次数: 0

2022

Aging-Aware Training for Printed Neuromorphic Circuits

2022 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD
Zhao, Haibin; Hefenbrock, Michael; Beigl, Michael; Tahoori, Mehdi B.

被引用次数: 5

2022

MVSTT: A Multi-Value Computation-in-Memory based on Spin-Transfer Torque Memories

2022 25TH EUROMICRO CONFERENCE ON DIGITAL SYSTEM DESIGN (DSD)
Jafari, Atousa; Mayahinia, Mahta; Ahmed, Soayed Tuhin; Muench, Christopher; Tahoori, Mehdi B.

被引用次数: 1

2022

Compact Functional Test Generation for Memristive Deep Learning Implementations using Approximate Gradient Ranking

2022 IEEE INTERNATIONAL TEST CONFERENCE (ITC)
Ahmed, Soayed Tuhin; Tahoori, Mehdi B.

被引用次数: 5

2022

A Survey of Test and Reliability Solutions for Magnetic Random Access Memories

PROCEEDINGS OF THE IEEE
Girard, Patrick; Cheng, Yuanqing; Virazel, Arnaud; Zhao, Weisheng; Bishnoi, Rajendra; Tahoori, Mehdi B.

被引用次数: 26

2021

Dynamic Behavior Predictions for Fast and Efficient Hybrid STT-MRAM Caches

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS
Sayed, Nour; Mao, Longfei; Tahoori, Mehdi B.

被引用次数: 6

2021

Voltage-Based Covert Channels Using FPGAs

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS
Gnad, Dennis R. E.; Nguyen, Cong Dang Khoa; Gillani, Syed Hashim; Tahoori, Mehdi B.

被引用次数: 9

2021

Low-Frequency Noise Characteristics of Inkjet-Printed Electrolyte-Gated Thin-Film Transistors

IEEE ELECTRON DEVICE LETTERS
Feng, Xiaowei; Singaraju, Surya Abhishek; Hu, Hongrong; Marques, Gabriel Cadilha; Fu, Tongtong; Baumgartner, Peter; Secker, Daniel; Tahoori, Mehdi B.; Aghassi-Hagmann, Jasmin

被引用次数: 0

2021

Defect Detection in Transparent Printed Electronics Using Learning-Based Optical Inspection

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Erozan, Ahmet Turan; Bosse, Simon; Tahoori, Mehdi B.

被引用次数: 3

2021

Channel Geometry Scaling Effect in Printed Inorganic Electrolyte-Gated Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rasheed, Farhan; Rommel, Manuel; Marques, Gabriel Cadilha; Wenzel, Wolfgang; Tahoori, Mehdi B.; Aghassi-Hagmann, Jasmin

被引用次数: 4

2021

High-Performance Radiation-Hardened Spintronic Retention Latch and Flip-Flop for Highly Reliable Processors

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Nair, Sarath Mohanachandran; Mayahinia, Mahta; Tahoori, Mehdi B.; Perumkunnil, Manu; Zahedmanesh, Houman; Croes, Kristof; Garelo, Kevin; Marinelli, Tommaso; Evenblij, Timon; Kar, Gouri Sankar; Catthoor, Francky

被引用次数: 5

2021

An Inside Job: Remote Power Analysis Attacks on FPGAs

IEEE DESIGN & TEST

Schellenberg, Falk; Gnad, Dennis R. E.; Moradi, Amir; Tahoori, Mehdi B.

被引用次数: 20

2021

Realization and training of an inverter-based printed neuromorphic computing system

SCIENTIFIC REPORTS

Weller, Dennis D.; Hefenbrock, Michael; Beigl, Michael; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 10

2021

Printed Low-Voltage Crossbar-PUF for Identification

2021 IEEE INTERNATIONAL FLEXIBLE ELECTRONICS TECHNOLOGY CONFERENCE (IFETC)

Scholz, Alexander; Zimmermann, Lukas; Sikora, Axel; Tahoori, Mehdi B.; Aghassi-Hagmann, Jasmin

被引用次数: 1

2021

TRIGON: A Single-phase-clocking Low Power Hardened Flip-Flop with Tolerance to Double-Node-Upset for Harsh Environments Applications

PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)

Li, Yan; Han, Jun; Zeng, Xiaoyang; Tahoori, Mehdi B.

被引用次数: 1

2021

Printed Stochastic Computing Neural Networks

PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)

Weller, Dennis D.; Bleier, Nathaniel; Hefenbrock, Michael; Aghassi-Hagmann, Jasmin; Beigl, Michael; Kumar, Rakesh; Tahoori, Mehdi B.

被引用次数: 11

2021

Stealthy Logic Misuse for Power Analysis Attacks in Multi-Tenant FPGAs

PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)

Gnad, Dennis R. E.; Meyers, Vincent; Dang, Nguyen Minh; Schellenberg, Falk; Moradi, Amir; Tahoori, Mehdi B.

被引用次数: 9

2021

Testing Resistive Memory based Neuromorphic Architectures using Reference Trimming

PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)

Muench, Christopher; Tahoori, Mehdi B.

被引用次数: 4

2021

Remote and Stealthy Fault Attacks on Virtualized FPGAs

PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)

Krautter, Jonas; Gnad, Dennis R. E.; Tahoori, Mehdi B.

被引用次数: 11

2021

Analyzing and Mitigating Sensing Failures in Spintronic-based Computing in Memory

2021 IEEE INTERNATIONAL TEST CONFERENCE (ITC 2021)

Mayahinia, Mahta; Muench, Christopher; Tahoori, Mehdi B.

被引用次数: 2

2021

Is your secure test infrastructure secure enough? Attacks based on delay test patterns using transient behavior analysis

2021 IEEE INTERNATIONAL TEST CONFERENCE (ITC 2021)

Meschkov, Sergej; Gnad, Dennis R. E.; Krautter, Jonas; Tahoori, Mehdi B.

被引用次数: 2

2021

LoopBreaker: Disabling Interconnects to Mitigate Voltage-Based Attacks in Multi-Tenant FPGAs

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Nassar, Hassan; AlZughbi, Hanna; Gnad, Dennis R. E.; Bauer, Lars; Tahoori, Mehdi B.; Henkel, Jorg

被引用次数: 7

2021

A Robust Hardened Latch Featuring Tolerance to Double-Node-Upset in 28nm CMOS for Spaceborne Application

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Li, Yan; Cheng, Xu; Tan, Chiyu; Han, Jun; Zhao, Yuanfu; Wang, Liang; Li, Tongde; Tahoori, Mehdi B.; Zeng, Xiaoyang

被引用次数: 23

2020

Reverse Engineering of Printed Electronics Circuits: From Imaging to Netlist Extraction

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Erozan, Ahmet Turan; Hefenbrock, Michael; Beigl, Michael; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 8

2020

Bayesian Optimized Mixture Importance Sampling for High-Sigma Failure Rate Estimation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Weller, Dennis D.; Hefenbrock, Michael; Golanbari, Mohammad S.; Beigl, Michael; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 3

2020

Selective Flip-Flop Optimization for Reliable Digital Circuit Design

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Golanbari, Mohammad Saber; Kiammehr, Saman; Ebrahimi, Mojtaba; Tahoori, Mehdi B.

被引用次数: 4

2020

Fabrication, Characterization and Simulation of Sputtered Pt/In-Ga-Zn-O Schottky Diodes for Low-Frequency Half-Wave Rectifier Circuits

IEEE ACCESS

Ulianova, Veronika; Rasheed, Farhan; Bolat, Sami; Sevilla, Galo Torres; Didenko, Yurii; Feng, Xiaowei; Shorubalko, Ivan; Bachmann, Dominik; Tatarchuk, Dmytro; Tahoori, Mehdi B.; Aghassi-Hagmann, Jasmin; Romanyuk, Yaroslav E.

被引用次数: 3

2020

Approximate Spintronic Memories

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Sayed, Nour; Bishnoi, Rajendra; Tahoori, Mehdi B.

被引用次数: 0

2020

Crossover-aware Placement and Routing for Inkjet Printed Circuits

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Rasheed, Farhan; Hefenbrock, Michael; Bishnoi, Rajendra; Beigl, Michael; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 1

2020

A Printed Camouflaged Cell Against Reverse Engineering of Printed Electronics Circuits

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Erozan, Ahmet Turan; Weller, Dennis D.; Feng, Yijing; Marques, Gabriel Cadilha; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 2

2020

A Compact Low-Voltage True Random Number Generator Based on Inkjet Printing Technology

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Erozan, Ahmet Turan; Wang, Guan Ying; Bishnoi, Rajendra; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 11

2020

A Novel Printed-Lookup-Table-Based Programmable Printed Digital Circuit

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Erozan, Ahmet Turan; Weller, Dennis D.; Rasheed, Farhan; Bishnoi, Rajendra; Aghassi-Hagmann, Jasmin; Tahoori, Mehdi B.

被引用次数: 3

2020

Achieving Energy Efficiency for Near-Threshold Circuits Through Postfabrication Calibration and Adaptation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Golanbari, Mohammad Saber; Kiamehr, Saman; Oboril, Fabian; Gebregiorgis, Anteneh; Tahoori, Mehdi B.

被引用次数: 3

2020

Secure STT-MRAM Bit-Cell Design Resilient to Differential Power Analysis Attacks

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ben Dodo, Samir; Bishnoi, Rajendra; Tahoori, Mehdi B.

被引用次数: 9

2020

An Inkjet-Printed Full-Wave Rectifier for Low-Voltage Operation Using Electrolyte-Gated Indium-Oxide Thin-Film Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

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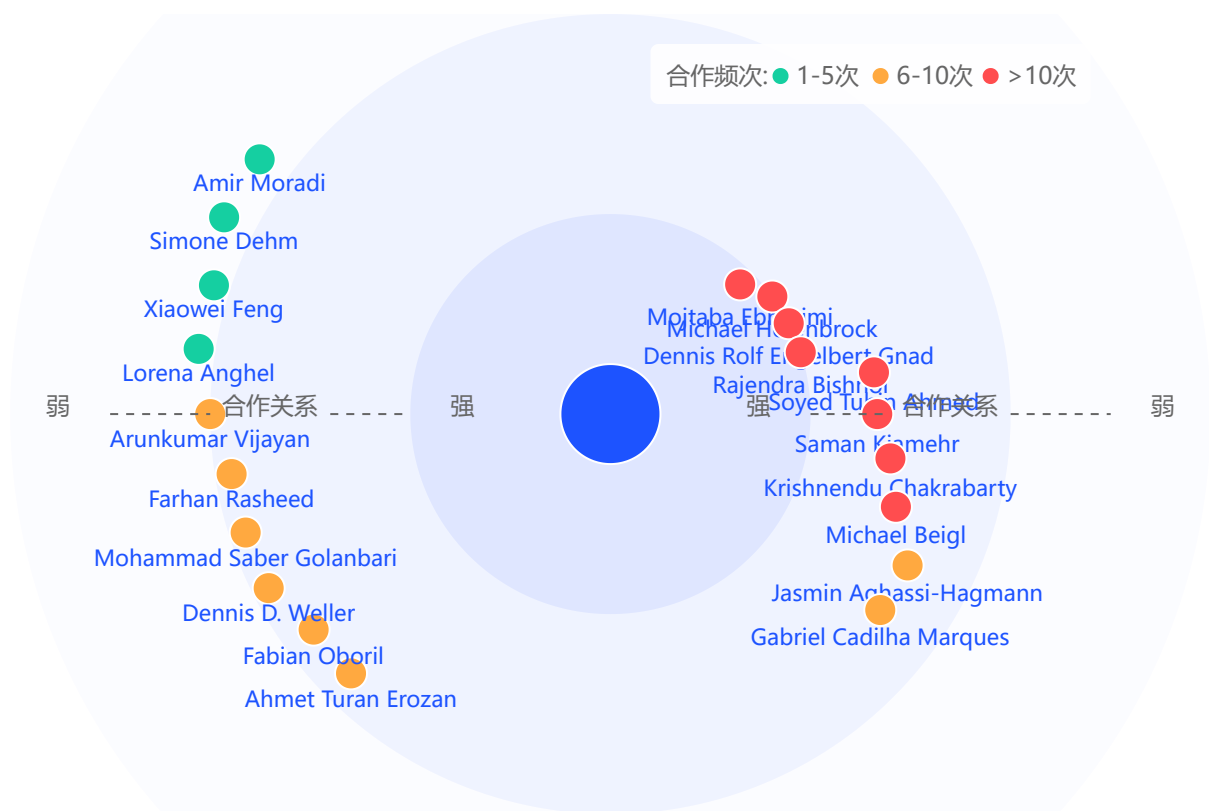
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